

FIGURE 9.5.3 Model of a hill in Example 7

EXAMPLE 8 Direction to Cool Off Fastest

The temperature in a rectangular box is approximated by
 $T(x, y, z) = xyz(1 - x)(2 - y)(3 - z)$, $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.

If a mosquito is located at $(\frac{1}{2}, 1, 1)$, in which direction should it fly to cool off as rapidly as possible?

SOLUTION

The gradient of T is

$$\nabla T(x, y, z) = yz(2 - y)(3 - z)(1 - 2x)\mathbf{i} + xz(1 - x)(3 - z)(2 - 2y)\mathbf{j} + xy(1 - x)(2 - y)(3 - 2z)\mathbf{k}.$$

Therefore, $\nabla T(\frac{1}{2}, 1, 1) = \frac{1}{4}\mathbf{k}$. To cool off most rapidly, the mosquito should fly in the direction of $-\frac{1}{4}\mathbf{k}$; that is, it should dive for the floor of the box, where the temperature is $T(x, y, 0) = 0$. ≡

9.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-24.

In Problems 1–4, compute the gradient for the given function.

1. $f(x, y) = x^2 - x^3y^2 + y^4$

2. $f(x, y) = y - e^{-2x^2y}$
3. $F(x, y, z) = \frac{xy^2}{z^3}$
4. $F(x, y, z) = xy \cos yz$

In Problems 5–8, find the gradient of the given function at the indicated point.

5. $f(x, y) = x^2 - 4y^2$; (2, 4)
6. $f(x, y) = \sqrt{x^3y - y^4}$; (3, 2)
7. $F(x, y, z) = x^2z^2 \sin 4y$; $(-2, \pi/3, 1)$
8. $F(x, y, z) = \ln(x^2 + y^2 + z^2)$; $(-4, 3, 5)$

In Problems 9 and 10, use Definition 9.5.1 to find $D_{\mathbf{u}} f(x, y)$ given that $\mathbf{u}$ makes the indicated angle with the positive x -axis.

9. $f(x, y) = x^2 + y^2$; $\theta = 30^\circ$
10. $f(x, y) = 3x - y^2$; $\theta = 45^\circ$

In Problems 11–20, find the directional derivative of the given function at the given point in the indicated direction.

11. $f(x, y) = 5x^3y^6$; $(-1, 1)$, $\theta = \pi/6$
12. $f(x, y) = 4x + xy^2 - 5y$; $(3, -1)$, $\theta = \pi/4$
13. $f(x, y) = \tan^{-1} \frac{y}{x}$; $(2, -2)$, $\mathbf{i} - 3\mathbf{j}$
14. $f(x, y) = \frac{xy}{x + y}$; $(2, -1)$, $6\mathbf{i} + 8\mathbf{j}$
15. $f(x, y) = (xy + 1)^2$; $(3, 2)$, in the direction of $(5, 3)$
16. $f(x, y) = x^2 \tan y$; $(\frac{1}{2}, \pi/3)$, in the direction of the negative x -axis
17. $F(x, y, z) = x^2y^2(2z + 1)^2$; $(1, -1, 1)$, $\langle 0, 3, 3 \rangle$
18. $F(x, y, z) = \frac{x^2 - y^2}{z^2}$; $(2, 4, -1)$, $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

- 19.** $F(x, y, z) = \sqrt{x^2y + 2y^2z}$; $(-2, 2, 1)$, in the direction of the negative z -axis
- 20.** $F(x, y, z) = 2x - y^2 + z^2$; $(4, -4, 2)$, in the direction of the origin

In Problems 21 and 22, consider the plane through the points P and Q that is perpendicular to the xy -plane. Find the slope of the tangent at the indicated point to the curve of intersection of this plane and the graph of the given function in the direction of Q .

- 21.** $f(x, y) = (x - y)^2$; $P(4, 2)$, $Q(0, 1)$; $(4, 2, 4)$
- 22.** $f(x, y) = x^3 - 5xy + y^2$; $P(1, 1)$, $Q(-1, 6)$; $(1, 1, -3)$

In Problems 23–26, find a vector that gives the direction in which the given function increases most rapidly at the indicated point. Find the maximum rate.

- 23.** $f(x, y) = e^{2x} \sin y$; $(0, \pi/4)$
- 24.** $f(x, y) = xye^{x-y}$; $(5, 5)$
- 25.** $F(x, y, z) = x^2 + 4xz + 2yz^2$; $(1, 2, -1)$
- 26.** $F(x, y, z) = xyz$; $(3, 1, -5)$

In Problems 27–30, find a vector that gives the direction in which the given function decreases most rapidly at the indicated point. Find the minimum rate.

- 27.** $f(x, y) = \tan(x^2 + y^2)$; $(\sqrt{\pi/6}, \sqrt{\pi/6})$
- 28.** $f(x, y) = x^3 - y^3$; $(2, -2)$
- 29.** $F(x, y, z) = \sqrt{xz}e^y$; $(16, 0, 9)$
- 30.** $F(x, y, z) = \ln\left(\frac{xy}{z}\right)$; $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$
- 31.** Find the directional derivative(s) of $f(x, y) = x + y^2$ at $(3, 4)$ in the direction of a tangent vector to the graph of $2x^2 + y^2 = 9$ at $(2, 1)$.

- 32.** If $f(x, y) = x^2 + xy + y^2 - x$, find all points where $D_{\mathbf{u}} f(x, y)$ in the direction of $\mathbf{u} = (1/\sqrt{2})(\mathbf{i} + \mathbf{j})$ is zero.
- 33.** Suppose $\nabla f(a, b) = 4\mathbf{i} + 3\mathbf{j}$. Find a unit vector $\mathbf{u}$ so that
- (a) $D_{\mathbf{u}} f(a, b) = 0$,
 - (b) $D_{\mathbf{u}} f(a, b)$ is a maximum, and
 - (c) $D_{\mathbf{u}} f(a, b)$ is a minimum.
- 34.** Suppose $D_{\mathbf{u}} f(a, b) = 6$. What is the value of $D_{-\mathbf{u}} f(a, b)$?
- 35.**

(a) If $f(x, y) = x^3 - 3x^2y^2 + y^3$, find the directional derivative of f at a point (x, y) in the direction of $\mathbf{u} = (1/\sqrt{10})(3\mathbf{i} + \mathbf{j})$.

(b) If $F(x, y) = D_{\mathbf{u}} f(x, y)$ in part (a), find $D_{\mathbf{u}} F(x, y)$.

- 36.** Consider the gravitational potential

$$U(x, y) = \frac{-Gm}{\sqrt{x^2 + y^2}},$$

where G and m are constants. Show that U increases or decreases most rapidly along a line through the origin.

- 37.** If $f(x, y) = x^3 - 12x + y^2 - 10y$, find all points at which $\|\nabla f\| = 0$.
- 38.** Suppose

$$D_{\mathbf{u}} f(a, b) = 7, D_{\mathbf{v}} f(a, b) = 3$$

$$\mathbf{u} = \frac{5}{13}\mathbf{i} - \frac{12}{13}\mathbf{j}, \quad \mathbf{v} = \frac{5}{13}\mathbf{i} + \frac{12}{13}\mathbf{j}.$$

Find $\nabla f(a, b)$.

- 39.** Consider the rectangular plate shown in **FIGURE 9.5.4**. The temperature at a point (x, y) on the plate is

given by $T(x, y) = 5 + 2x^2 + y^2$. Determine the direction an insect should take, starting at $(4, 2)$, in order to cool off as rapidly as possible.

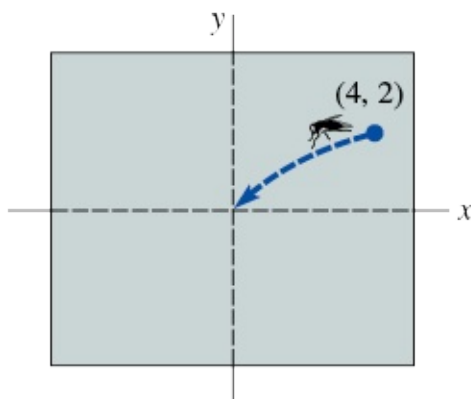


FIGURE 9.5.4 Insect in Problem 39

- 40.** In Problem 39, observe that $(0, 0)$ is the coolest point of the plate. Find the path the cold-seeking insect, starting at $(4, 2)$, will take to the origin. If $\langle x(t), y(t) \rangle$ is the vector equation of the path, then use the fact that $-\nabla T(x, y) = \langle x'(t), y'(t) \rangle$. Why is this? [*Hint: Remember separation of variables?*]
- 41.** The temperature at a point (x, y) on a rectangular metal plate is given by $T(x, y) = 100 - 2x^2 - y^2$. Find the path a heat-seeking particle will take, starting at $(3, 4)$, as it moves in the direction in which the temperature increases most rapidly.
- 42.** The temperature T at a point (x, y, z) in space is inversely proportional to the square of the distance from (x, y, z) to the origin. It is known that $T(0, 0, 1) = 500$. Find the rate of change of T at $(2, 3, 3)$ in the direction of $(3, 1, 1)$. In which direction from $(2, 3, 3)$ does the temperature T increase most rapidly? At $(2, 3, 3)$ what is the maximum rate of change of T ?
- 43.** Find a function f such that

$$\nabla f = (3x^2 + y^3 + ye^{xy})\mathbf{i} + (-2y^2 + 3xy^2 + xe^{xy})\mathbf{j}.$$

44. Let f_x, f_y, f_{xy}, f_{yx} be continuous and $\mathbf{u}$ and $\mathbf{v}$ be unit vectors. Show that $D_{\mathbf{u}}D_{\mathbf{v}}f = D_{\mathbf{v}}D_{\mathbf{u}}f$.

In Problems 45–48, assume that f and g are differentiable functions of two variables. Prove the given identity.

45. $\nabla(cf) = c \nabla f$

46. $\nabla(f + g) = \nabla f + \nabla g$

47. $\nabla(fg) = f \nabla g + g \nabla f$

48. $\nabla\left(\frac{f}{g}\right) = \frac{g \nabla f - f \nabla g}{g^2}$

49. If $\mathbf{F}(x, y, z) = f_1(x, y, z)\mathbf{i} + f_2(x, y, z)\mathbf{j} + f_3(x, y, z)\mathbf{k}$ and

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z},$$

show that

$$\nabla \times \mathbf{F} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}$$

* Note that the numerator in (4) can be written $f(x + h \cos \alpha, y + h \cos \beta) - f(x, y)$, where $\beta = (\pi/2) - \alpha$.